

Multi-Horseshoe Dense Property and Intermediate Entropy Property of Ergodic Measures with Same Level

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- X : a compact metric space
- f : a continuous map on X
- $\mathcal{M}_f(X)$: set of f -invariant Borel probability measures (compact convex)
- $\mathcal{M}_f^e(X)$: subset of f -ergodic measures $= \text{ext}(\mathcal{M}_f(X))$.

Birkhoff ergodic theorem

Assume that ν is ergodic, then ν -almost every point x is ν -generic, meaning that

$$\frac{1}{n} \sum_{k=0}^{n-1} \delta_{f^k(x)} \rightarrow \nu$$

in weak* topology.

Entropy of generic points (Bowen 1973)

Assume that ν is ergodic, then the set of ν -generic points G_μ satisfies

$$h_{top}(f, G_\mu) = h_\mu(f),$$

where $h_{top}(f, G_\mu)$ is Bowen topological entropy of G_μ , $h_\mu(f)$ is the metric entropy of μ .

Lyapunov exponents of generic points (Fan-Wu 2022)

Let A be a continuous function on a subshift Σ in $\{0, 1, \dots, m-1\}^{\mathbb{N}}$ taking $d \times d$ non-negative matrices as values and let ν be an ergodic measure on Σ . Under some positivity conditions of A , for any ν -generic point ω , the Lyapunov exponent

$$L(\omega) := \lim_{n \rightarrow \infty} \frac{1}{n} \log \|A(\omega)A(\sigma\omega) \cdots A(\sigma^{n-1}\omega)\|$$

exists.

'proportion' of ergodic measures

Problem

To describe 'proportion' of ergodic measures in the set of invariant measures.

viewpoints

- 1 entropy
- 2 topology
- 3 Birkhoff average
- 4 emergence
- 5 universality
- 6 ...

Viewpoint of entropy: Katok's conjecture

Katok proved the following milestone theorem.

Theorem 2.1 (Katok 1980)

Let f be a C^r ($r > 1$) diffeomorphism on a surface M and μ be an f -ergodic measure with $h_\mu(f) > 0$. Then for any $\varepsilon > 0$, there exists a hyperbolic horseshoe Λ_ε satisfying

$$h_{\text{top}}(f, \Lambda_\varepsilon) > h_\mu(f) - \varepsilon.$$

From Theorem 2.1, every C^r ($r > 1$) surface diffeomorphism f has the intermediate entropy property, i.e., for any $h \in [0, h_{\text{top}}(f))$, there exists an f -ergodic measure μ satisfying $h_\mu(f) = h$. Consequently, the following two sets are equal:

$$\begin{aligned} & \{h_\mu(f) : \mu \text{ is an } f\text{-ergodic measure}\} \\ & \{h_\mu(f) : \mu \text{ is an } f\text{-invariant measure}\}. \end{aligned}$$

Katok believed that this holds in any dimension.

Conjecture (Katok)

For every C^2 diffeomorphism f on a Riemannian manifold M , the set

$$\{h_\mu(f) : \mu \text{ is an } f\text{-ergodic measure}\}$$

includes $[0, h_{top}(f))$.

Katok's conjecture

Progress on Katok's conjecture:

- ① certain skew products and toral automorphisms: Sun 2010, 2012
- ② some partially hyperbolic diffeomorphisms with one-dimensional center bundles: Ures 2012, Yang-Zhang 2021
- ③ asymptotic entropy expansiveness+almost weak specification property+small boundary property: Quas-Soo 2016
- ④ certain homogeneous systems: Guan-Sun-Wu 2017
- ⑤ hereditary shift: Konieczny-Kupsa-Kwietniak 2018
- ⑥ asymptotic entropy expansiveness+transitive+shadowing: Li-Oprocha 2018
- ⑦ asymptotic entropy expansiveness+approximate product property: Sun 2019
- ⑧ star flow: Li-Shi-Wang-Wang 2020

Problem

Under what conditions, is $\mathcal{M}_f^e(X)$ residual in $\mathcal{M}_f(X)$?

Since $\mathcal{M}_f^e(X)$ is the set of extreme points of $\mathcal{M}_f(X)$, then $\mathcal{M}_f^e(X)$ is a G_δ subset of $\mathcal{M}_f(X)$. Thus, it's enough to prove $\mathcal{M}_f^e(X)$ is dense in $\mathcal{M}_f(X)$.

Theorem 2.2 (Sigmund 1970)

If (X, f) satisfies the specification property, then $\mathcal{M}_f^e(X)$ is residual in $\mathcal{M}_f(X)$. If further the entropy function is upper semi-continuous, then $\{\mu \in \mathcal{M}_f^e(X) : h_\mu(f) = 0\}$ is residual in $\mathcal{M}_f(X)$.

Theorem 2.3 (Eizenberg-Kifer-Weiss 1994)

If (X, f) satisfies the specification property, then for any $\mu \in \mathcal{M}_f(X)$ there exists $\{\mu_n\}_{n=1}^{\infty} \subset \mathcal{M}_f^e(X)$ such that

$$\lim_{n \rightarrow \infty} \mu_n = \mu \text{ and } \lim_{n \rightarrow \infty} h_{\mu_n}(f) \geq h_{\mu}(f).$$

Pfister and Sullivan referred to this property as the entropy-dense property and proved that the entropy-dense property holds for systems with the approximate product property.

Theorem 2.4 (Li-Oprocha 2018)

If (X, f) is a transitive dynamical system with the shadowing property, for any $\mu \in \mathcal{M}_f(X)$ and every $0 \leq c \leq h_\mu(f)$, there exists $\{\mu_n\}_{n=1}^\infty \subset \mathcal{M}_f^e(X)$ such that

$$\lim_{n \rightarrow \infty} \mu_n = \mu \text{ and } \lim_{n \rightarrow \infty} h_{\mu_n}(f) = c.$$

If further the entropy function is upper semi-continuous, then for every $0 \leq c < h_{\text{top}}(f)$, the set of ergodic measures with entropy c is residual in the space of invariant measures with entropy at least c .

Theorem 2.5 (Tian-Wang-Wang 2019)

Suppose that (X, f) has the periodic gluing orbit property. Let $\varphi : X \rightarrow \mathbb{R}$ be a continuous function with

$$\inf_{\mu \in \mathcal{M}_f(X)} \int \varphi d\mu < \sup_{\mu \in \mathcal{M}_f(X)} \int \varphi d\mu.$$

Then, for any

$$\inf_{\mu \in \mathcal{M}_f(X)} \int \varphi d\mu < a < \sup_{\mu \in \mathcal{M}_f(X)} \int \varphi d\mu,$$

there is an ergodic measure μ_a such that $\int \varphi d\mu_a = a$.

Main results

Let f be a diffeomorphism on a Riemannian manifold M . Denote $C(M, \mathbb{R})$ the set of continuous function on M . We would like to consider the following question.

Question

For every diffeomorphism f on a Riemannian manifold M and every $\varphi \in C(M, \mathbb{R})$, whether the following two sets are equal

$$\begin{aligned} & \left\{ \left(\int \varphi d\mu, h_\mu(f) \right) : \mu \in \mathcal{M}_f^e(M) \right\}, \\ & \left\{ \left(\int \varphi d\mu, h_\mu(f) \right) : \mu \in \mathcal{M}_f(M) \right\}? \end{aligned}$$

Moreover, given $(a, h) \in \left\{ \left(\int \varphi d\mu, h_\mu(f) \right) : \mu \in \mathcal{M}_f(X) \right\}$, is

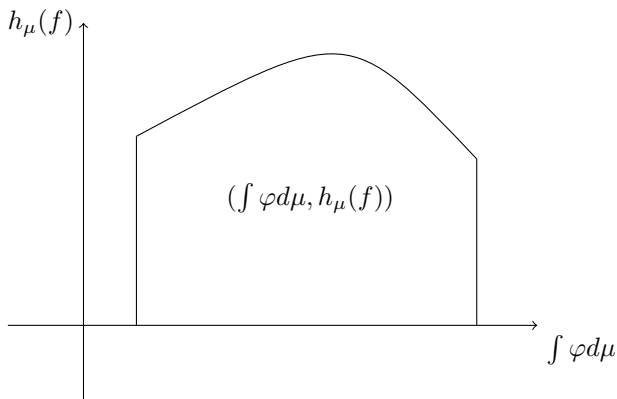
$$\left\{ \mu \in \mathcal{M}_f^e(X) : \left(\int \varphi d\mu, h_\mu(f) \right) = (a, h) \right\}$$

residual in

$$\left\{ \mu \in \mathcal{M}_f(X) : \left(\int \varphi d\mu, h_\mu(f) \right) = (a, h) \right\}?$$

Main results

Figure: Graph of $(\int \varphi d\mu, h_\mu(f))$



Theorem A (Dong-H.-Tian 2022)

Let f be a diffeomorphism on a Riemannian manifold M and $X \subseteq M$ be a transitive locally maximal hyperbolic set of f . If $\varphi \in C(X, \mathbb{R})$, then the following two sets are equal:

$$\begin{aligned} & \text{Int}(\{(\int \varphi d\mu, h_\mu(f)) : \mu \in \mathcal{M}_f(X)\}), \\ & \text{Int}(\{(\int \varphi d\mu, h_\mu(f)) : \mu \in \mathcal{M}_f^e(X)\}). \end{aligned}$$

Moreover, given $(a, h) \in \text{Int}(\{(\int \varphi d\mu, h_\mu(f)) : \mu \in \mathcal{M}_f(X)\})$,

$$\{\mu \in \mathcal{M}_f^e(X) : (\int \varphi d\mu, h_\mu(f)) = (a, h)\}$$

is residual in

$$\{\mu \in \mathcal{M}_f(X) : (\int \varphi d\mu, h_\mu(f)) = (a, h)\}.$$

Corollary A

Let f be a diffeomorphism on a Riemannian manifold M and $X \subseteq M$ be a transitive locally maximal hyperbolic set of f . If φ is Hölder, then the following two sets are equal:

$$\begin{aligned} & \left\{ \left(\int \varphi d\mu, h_\mu(f) \right) : \mu \in \mathcal{M}_f(X), \int \varphi d\mu \in \text{Int}(L_\varphi) \right\}, \\ & \left\{ \left(\int \varphi d\mu, h_\mu(f) \right) : \mu \in \mathcal{M}_f^e(X), \int \varphi d\mu \in \text{Int}(L_\varphi) \right\}. \end{aligned}$$

Remark 3.1

Theorem A also is true if (X, f) is one of the following systems:

- ① transitive subshifts of finite type,
- ② transitive locally maximal hyperbolic set of flow,
- ③ generically geometric Lorenz attractor or singular hyperbolic attractor,
- ④ β -shifts,
- ⑤ S -gap shifts.

Remark 3.2

We replace $\int \varphi d\mu$ by $\left(\frac{\lim_{n \rightarrow \infty} \frac{1}{n} \int \varphi_n^1 d\mu}{\lim_{n \rightarrow \infty} \frac{1}{n} \int \psi_n^1 d\mu}, \dots, \frac{\lim_{n \rightarrow \infty} \frac{1}{n} \int \varphi_n^d d\mu}{\lim_{n \rightarrow \infty} \frac{1}{n} \int \psi_n^d d\mu} \right)$ which satisfies

$$\liminf_{m \rightarrow \infty} \frac{\psi_m^i(x)}{m} > 0 \quad \text{and} \quad \psi_n^i(x) > 0$$

for every $i = 1, \dots, d, x \in X$, and $n \in \mathbb{N}$,

- ① if each $(\varphi_n^i)_{n \in \mathbb{N}}$ and $(\psi_n^i)_{n \in \mathbb{N}}$ are asymptotically additive, then Theorem A also holds.
- ② if each $(\varphi_n^i)_{n \in \mathbb{N}}$ and $(\psi_n^i)_{n \in \mathbb{N}}$ are almost additive and have bounded variation, then Corollary A also holds.

Conditional variational principle

Given $\varphi \in C(X, \mathbb{R})$. Denote

$$L_\varphi(f, X) = \left[\inf_{\mu \in \mathcal{M}_f(X)} \int \varphi d\mu, \sup_{\mu \in \mathcal{M}_f(X)} \int \varphi d\mu \right].$$

For any $a \in L_\varphi(f, X)$, set

$$R_\varphi(a) := \{x \in X : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \varphi(f^i(x)) = a\}.$$

Conditional variational principle

Barreira and Saussol established the following conditional variational principle.

Theorem 4.1 (Barreira-Saussol 2001)

Suppose (X, f) is a dynamical system whose entropy function is upper semi-continuous. If $\varphi \in C(X, \mathbb{R})$ has a unique equilibrium measure, then for any $a \in \text{Int}(L_\varphi(f, X))$, one has $R_\varphi(a) \neq \emptyset$ and the following properties hold:

- ① $h_{\text{top}}(f, R_\varphi(a))$ satisfies the variational principle:

$$h_{\text{top}}(f, R_\varphi(a)) = \sup \left\{ h_\mu(f) : \mu \in \mathcal{M}_f(X) \text{ and } \int \varphi d\mu = a \right\}.$$

- ② There is $\mu_a \in \mathcal{M}_f^e(X)$ with $\int \varphi d\mu_a = a$ such that

$$h_{\text{top}}(f, R_\varphi(a)) = h_{\mu_a}(f).$$

Conditional variational principle

For continuous functions, Climenhaga obtained the following conditional variational principle.

Theorem 4.2 (Climenhaga 2013)

Suppose (X, f) is a dynamical system whose entropy function is upper semi-continuous and $h_{\text{top}}(f, X) < \infty$. Assume that there is a dense subspace D of $C(X, \mathbb{R})$ such that every $\psi \in D$ has a unique equilibrium measure. Let $\varphi \in C(X, \mathbb{R})$. If $a \in \text{Int}(L_\varphi(f, X))$, then $R_\varphi(a) \neq \emptyset$ and the following properties hold:

- ① $h_{\text{top}}(f, R_\varphi(a))$ satisfies the variational principle:

$$h_{\text{top}}(f, R_\varphi(a)) = \sup \left\{ h_\mu(f) : \mu \in \mathcal{M}_f(X) \text{ and } \int \varphi d\mu = a \right\}.$$

- ② For any $\varepsilon > 0$, there is $\mu_a \in \mathcal{M}_f^e(X)$ with $\int \varphi d\mu_a = a$ such that

$$|h_{\text{top}}(f, R_\varphi(a)) - h_{\mu_a}(f)| < \varepsilon.$$

Uniqueness of equilibrium measure

Theorem 4.3 (Bowen 1974)

Let f be a diffeomorphism on a Riemannian manifold M and $X \subseteq M$ be a transitive locally maximal hyperbolic set of f . Then every Hölder continuous function on X has a unique equilibrium measure.

Let $D(X)$ be the space of Hölder continuous functions on X . Then $D(X)$ is dense in $C(X, \mathbb{R})$. So the result of Theorem 4.2 holds for transitive locally maximal hyperbolic sets.

'Multi-horseshoe' entropy-dense property

For topological dynamical systems we introduce 'multi-horseshoe' entropy-dense property and show that it holds for topologically transitive Anosov system.

Definition 5.1

A homeomorphism $f : X \rightarrow X$ of a compact metric space called topologically Anosov, if (X, f) is expansive and satisfies the shadowing property.

'Multi-Horseshoe' Entropy-Dense Property

For any $m \in \mathbb{N}$ and $\{\nu_i\}_{i=1}^m \subseteq \mathcal{M}_f(X)$, we write $\text{cov}\{\nu_i\}_{i=1}^m$ for the convex combination of $\{\nu_i\}_{i=1}^m$, namely,

$$\text{cov}\{\nu_i\}_{i=1}^m := \left\{ \sum_{i=1}^m t_i \nu_i : t_i \in [0, 1], 1 \leq i \leq m \text{ and } \sum_{i=1}^m t_i = 1 \right\}.$$

Theorem 5.2 (Dong-H.-Tian 2022)

Suppose (X, f) is transitive and topologically Anosov. Then (X, f) satisfies the 'multi-horseshoe' entropy-dense property. That is, for any positive integer m , any f -invariant measures $\{\mu_i\}_{i=1}^m$ and any $\eta, \zeta > 0$, there exist compact invariant subsets $\Lambda_i \subseteq \Lambda \subsetneq X$ such that for each $1 \leq i \leq m$

- ① (Λ_i, f) and (Λ, f) conjugate to transitive subshifts of finite type,
- ② $h_{\text{top}}(f, \Lambda_i) > h_{\mu_i}(f) - \eta$,
- ③ $d_H(\text{cov}\{\mu_i\}_{i=1}^m, \mathcal{M}_f(\Lambda)) < \zeta$, $d_H(\mu_i, \mathcal{M}_f(\Lambda_i)) < \zeta$.

Proof of Theorem A:

Let f be a diffeomorphism on a Riemannian manifold M and $X \subseteq M$ be a transitive locally maximal hyperbolic set of f . Fix $\varphi \in C(X, \mathbb{R})$.

Lemma 6.1

For any $a \in \text{Int}(L_\varphi(f, X))$, any $\mu_0 \in \mathcal{M}_f(X)$ with $\int \varphi d\mu_0 = a$ and any $\eta, \zeta > 0$, there is $\nu \in \mathcal{M}_f^e(X)$ with $\int \varphi d\nu = a$ such that $\rho(\nu, \mu_0) < \zeta$ and $|h_\nu(f) - h_{\mu_0}(f)| < \eta$.

Proof of Lemma 6.1: Fix $a \in \text{Int}(L_\varphi(f, X))$, $\mu_0 \in \mathcal{M}_f(X)$ with $\int \varphi d\mu_0 = a$ and $\eta, \zeta > 0$. Since the entropy function is upper semi-continuous, there is $0 < \zeta' < \zeta$ such that

$$h_\omega(f) < h_{\mu_0}(f) + \frac{3\eta}{4}. \quad (6.1)$$

for any $\omega \in \mathcal{M}_f(X)$ with $\rho(\mu_0, \omega) < \zeta'$. Since $a \in \text{Int}(L_\varphi(f, X))$, there exist $\mu_1, \mu_2 \in \mathcal{M}_f(X)$ such that

$$h_{\mu_i}(f) > h_{\mu_0}(f) - \frac{\eta}{4}, \quad \rho(\mu_i, \mu_0) < \frac{\zeta'}{2} \text{ for each } i \in \{1, 2\}, \quad (6.2)$$

and

$$\int \varphi d\mu_1 > a, \quad \int \varphi d\mu_2 < a. \quad (6.3)$$

There is $0 < \zeta'' < \zeta'$ such that

$$\int \varphi d\omega_1 > a, \quad \int \varphi d\omega_2 < a. \quad (6.4)$$

for any $\omega_1, \omega_2 \in \mathcal{M}_f(X)$ with $\rho(\omega_1, \mu_1) < \zeta''$ and $\rho(\omega_2, \mu_2) < \zeta''$.

For the two invariant measures μ_1, μ_2 , there exist compact invariant subsets $\Lambda_1, \Lambda_2 \subseteq \Lambda \subsetneq X$ such that for each $i \in \{1, 2\}$

- ① (Λ_i, f) and (Λ, f) conjugate to transitive subshifts of finite type,
- ② $h_{\text{top}}(f, \Lambda_i) > h_{\mu_i}(f) - \frac{\eta}{4}$,
- ③ $d_H(\text{cov}\{\mu_i\}_{i=1}^2, \mathcal{M}_f(\Lambda)) < \frac{\xi''}{2}$, $d_H(\mu_i, \mathcal{M}_f(\Lambda_i)) < \frac{\xi''}{2}$.

By item 2, there are $\nu_i \in \mathcal{M}_f(\Lambda_i)$ such that

$$h_{\nu_i}(f) > h_{\text{top}}(f, \Lambda_i) - \frac{\eta}{4} > h_{\mu_i}(f) - \frac{2\eta}{4} > h_{\mu_0}(f) - \frac{3\eta}{4}.$$

By item 2 and (6. 4), we have

$$\int \varphi d\nu_1 > a, \int \varphi d\nu_2 < a. \quad (6. 5)$$

Then there is $\theta \in (0, 1)$ such that $\int \varphi d\mu' = a$ where $\nu' = \theta\nu_1 + (1 - \theta)\nu_2$. Thus on one hand we have

$$\begin{aligned} H(f, a, \Lambda) &:= \sup \left\{ h_{\mu}(f) : \mu \in \mathcal{M}_f(\Lambda) \text{ and } \int \varphi d\mu = a \right\} \\ &\geq h_{\nu'}(f) \geq \min\{h_{\nu_1}(f), h_{\nu_2}(f)\} > h_{\mu_0}(f) - \frac{3\eta}{4}. \end{aligned} \quad (6. 6)$$

On the other hand, by item 3, (6. 2) and (6. 1) we have

$$H(f, a, \Lambda) < h_{\mu_0}(f) + \frac{3\eta}{4}. \quad (6. 7)$$

From [Anosov 2010] any hyperbolic set conjugate to a subshift of finite type is locally maximal. Thus Λ is a transitive locally maximal hyperbolic set. So there is $\nu \in \mathcal{M}_f^e(\Lambda)$ with $\int \varphi d\nu = a$ such that $|h_\nu(f) - H(f, a, \Lambda)| < \frac{\eta}{4}$. It follows that $|h_\nu(f) - h_{\mu_0}(f)| < \eta$. By item 2 and (6. 2) we have $\rho(\nu, \mu_0) < \zeta' < \zeta$. $\square$

Lemma 6.2

For any $a \in \text{Int}(L_\varphi(f, X))$, any $\mu_0 \in \mathcal{M}_f(X)$ with $\int \varphi d\mu_0 = a$, any $0 \leq c \leq h_{\mu_0}(f)$ and any $\eta, \zeta > 0$, there is $\nu \in \mathcal{M}_f^e(X)$ with $\int \varphi d\nu = a$ such that $\rho(\nu, \mu_0) < \zeta$ and $|h_\nu(f) - c| < \eta$.

Proof of Lemma 6.2: Fix $a \in \text{Int}(L_\varphi(f, X))$, $\mu_0 \in \mathcal{M}_f(X)$ with $\int \varphi d\mu_0 = a$, $0 \leq c \leq h_{\mu_0}(f)$ and $\eta, \zeta > 0$. Since the set of periodic measures is dense in $\mathcal{M}_f(X)$, then $\{\mu \in \mathcal{M}_f(X) : h_\mu(f) = 0\}$ is dense in $\mathcal{M}_f(X)$, and thus there exists $\nu' \in \mathcal{M}_f(X)$ with $\int \varphi d\nu' = a$ such that $h_{\nu'}(f) = c$ and $\rho(\nu', \mu_0) < \frac{\zeta}{2}$. So by Lemma 6.1 there is $\nu \in \mathcal{M}_f(X)$ with $\int \varphi d\nu = a$ such that $\rho(\nu, \nu') < \frac{\zeta}{2}$ and $|h_\nu(f) - h_{\nu'}(f)| < \eta$. $\square$

Lemma 6.3

For any $a \in \text{Int}(L_\varphi(f, X))$, and $0 \leq c < \sup\{h_\mu(f) : \mu \in \mathcal{M}_f(\Lambda) \text{ and } \int \varphi d\mu = a\}$, the set $\{\mu \in \mathcal{M}_f^e(X) : h_\mu(f) = c \text{ and } \int \varphi d\mu = a\}$ is residual in $\{\mu \in \mathcal{M}_f(X) : h_\mu(f) \geq c \text{ and } \int \varphi d\mu = a\}$.

Proof of Lemma 6.3: From Lemma 6.2,

$\{\mu \in \mathcal{M}_f^e(X) : h_\mu(f) \geq c \text{ and } \int \varphi = a\}$ is dense in

$\{\mu \in \mathcal{M}_f(X) : h_\mu(f) \geq c \text{ and } \int \varphi = a\}$.

Since $\mathcal{M}_f^e(X)$ is a G_δ subset of $\mathcal{M}_f(X)$, then

$\{\mu \in \mathcal{M}_f^e(X) : h_\mu(f) \geq c \text{ and } \int \varphi d\mu = a\}$ is residual in

$\{\mu \in \mathcal{M}_f(X) : h_\mu(f) \geq c \text{ and } \int \varphi d\mu = a\}$.

Since the entropy function is upper semi-continuous, then

$\{\mu \in \mathcal{M}_f(X) : h_\mu(f) = c \text{ and } \int \varphi d\mu = a\}$ is residual in

$\{\mu \in \mathcal{M}_f(X) : h_\mu(f) \geq c \text{ and } \int \varphi d\mu = a\}$, and thus we complete the proof. □

By Lemma 6.3, then the following two sets are equal:

$$\begin{aligned} & \text{Int}(\{(\int \varphi d\mu, h_\mu(f)) : \mu \in \mathcal{M}_f(X)\}), \\ & \text{Int}(\{(\int \varphi d\mu, h_\mu(f)) : \mu \in \mathcal{M}_f^e(X)\}). \end{aligned}$$

If further φ is Hölder, then by Lemma 4.1 the following two sets are equal:

$$\begin{aligned} & \{(\int \varphi d\mu, h_\mu(f)) : \mu \in \mathcal{M}_f(X), \int \varphi d\mu \in \text{Int}(L_\varphi)\}, \\ & \{(\int \varphi d\mu, h_\mu(f)) : \mu \in \mathcal{M}_f^e(X), \int \varphi d\mu \in \text{Int}(L_\varphi)\}. \end{aligned}$$



Thank You!